Fundamentals of Manufacturing

Manufacturing Overview

- Manufacturing may be defined as the production of products by hand or by machine for sale or service. It requires the respective raw materials, the necessary tools and machines, and manpower.
- In the designing stage, products should be designed either for assembly (DFA), or for manufacture (DFM). This module focuses more on DFM; on how the product should be designed to best fit specific manufacturing methods.
- Generally, the following are considered when choosing manufacturing processesL
  - Geometry (Size, shape, form)
    - different manufacturing processes are better at producing different forms, like machining, casting, welding, and additive manufacturing
  - Properties & Characteristics
    - different mechanical properties needed requires specific processes, like heat treatment, surface hardening, or using a different material.
  - Surface Finish & Tolerances
    - if the part needs to fit or mate with other parts, specific surface finish and tolerance must be maintained, which can be achieved with processes like lapping and polishing
  - Cost
    - different processes, material choices and design choices influence the cost
- The manufacturing process can be thought of in two perspectives. Technologically, a manufacturing process involves introducing mechanical and chemical processes to a stock material to alter its geometry and improve its properties and appearance. To do so, it involves a combination of machinery, tools, power and labor, for both manufacturing of parts and assembly, if applicable.

From an economic perspective, the process helps to increase the utility and hence the value of the stock material, and can be thought of as a process that does so.

Engineering Materials & Material Selection

For engineering applications, there are 4 main classes of materials used, metals, plastics, ceramics and composites. However, mostly metals are used for applications.

Ferrous Metals & Alloys

Mainly contains iron, with alloying elements like carbon, and can have alloying element proportion varied.

Typically used for sheet metal, plates, structural elements, tooling, rods, and wires.

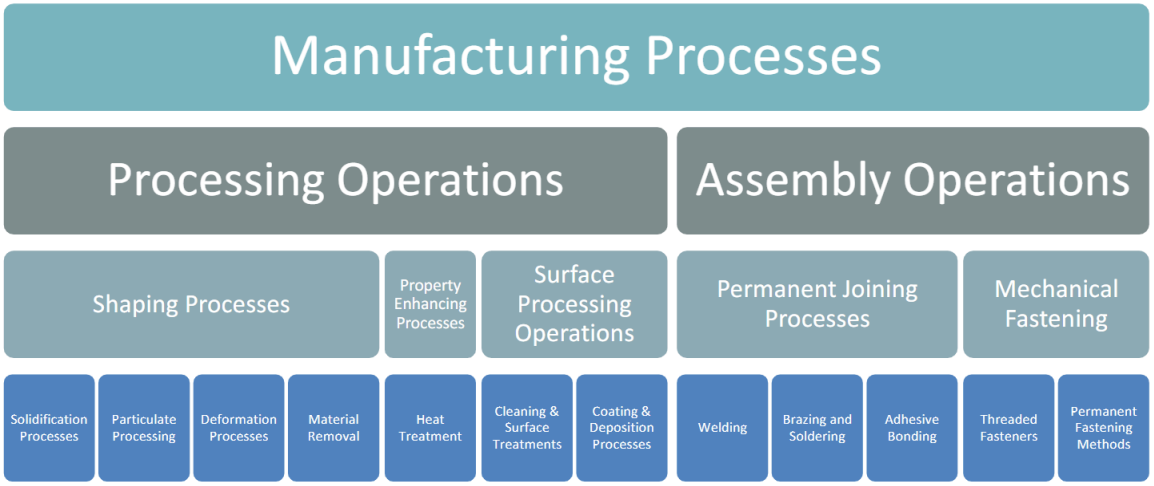
Non-ferrous Metals & Alloys

includes aluminium, copper, brass, zinc, titanium etc.

has good corrosion resistance and high thermal and electrical conductivity with low density, while having high east of production.

Manufacturing Methods & Selection

Shown are the classification of some common manufacturing processes:



But the most useful one are shaping processes.

Solidification Processes

essentially casting/moulding, basically any process that cools a material in the liquid state in a mould.

encapsulates injection moulding, sand casting, centrifugal casting, investment casting, etc.

For plastics, polymer processing methods have to be used instead, typically with melting/partially melting plastic pellets or tubing into the respective design, however the actual process depends on the purpose and the material type.

Particulate Processing

fine metal or ceramic powers are compressed into the desired shape with processes like uniaxial or isostatic pressing, before being sintered to bond the particles together.

Deformation Processes

involves bending, shaping and deforming the stock by applying forces above the yield strength to plastically deform the metal. usually heated below m.p. to improve ductility. involves processes like cold working, cold rolling, die extrusion and forging.

Material Removal

general subtractive machining methods like turning, milling, grinding, wire EDM, etc.

surface finish methods like polishing and lapping do count too.

PRIMA Selection Matrix

One way to decide the manufacturing method used is with the PRIMA selection matrix, which takes into account the material type and the quantity of parts required, to give a range of possible methods to be used:

MATERIAL / QUANTITY	IRONS	STEEL (carbon)	STEEL (tool, alloy)	STAINLESS STEEL	COPPER & ALLOYS	ALUMINIUM & ALLOYS	MAGNESIUM & ALLOYS	ZINC & ALLOYS	TIN & ALLOYS	LEAD & ALLOYS	NICKEL & ALLOYS	TITANIUM & ALLOYS	THERMOPLASTICS	THERMOSETS	FR COMPOSITES	CERAMICS	REFRACTORY METALS	PRECIOUS METALS
VERY LOW 1 TO 100	[3.5][3.6] [3.7][6M] [7.1][7.4]	[3.5][3.7] [4.10] [6M][7.1] [7.5][7.6]	[3.1][3.5] [3.7][4.10] [6M][7.1] [7.5][7.6][7.7]	[3.5][3.7] [4.7][4.10] [6M][7.1] [7.5][7.6]	[3.5][3.7] [4.10][6M] [7.1]	[3.5][3.7] [4.7][4.10] [6M][7.5]	[3.6][3.7] [4.10][6M] [7.1][7.5]	[3.1][3.7] [4.10][6M] [7.5]	[3.1][3.7] [4.10][6M] [7.5]	[3.1][4.10] [6M][7.5]	[3.5][3.7] [4.10][6M] [7.1][7.5][7.6]	[3.1][3.6][4.7] [4.10][6M] [7.1][7.5] [7.6][7.7]	[5.5] [5.7]	[5.5] [7.7]	[5.2] [5.8] [7.7]	[3.5] [7.1] [7.5] [7.6] [7.7]	[3.1] [7.7]	[7.5]
LOW 100 TO 1,000	[3.2][3.5] [3.6][3.7] [6M][7.1] [7.3][7.4]	[3.2][3.5] [3.7][4.10] [6M][7.1] [7.3][7.4] [7.5]	[3.1][3.2] [3.7][6M][7.1] [7.3][7.4][7.5] [7.6][7.7]	[3.2][3.7] [4.7][4.10] [6M][7.1] [7.3][7.4][7.5]	[3.2][3.5][3.7] [3.8][4.5] [4.10][6M] [7.1][7.3][7.5]	[3.2][3.5][3.7] [3.8][4.7] [4.10][6M] [7.3][7.4][7.5]	[3.6][3.7] [3.8][4.10] [6M][7.5]	[3.1][3.7] [3.8][4.10] [6M][7.5]	[3.1][3.7] [3.8][4.10] [6M][7.5]	[3.1][3.8] [4.10][6M] [7.5]	[3.2][3.5][3.7] [4.10][6M] [7.1][7.3] [7.4][7.5]	[3.1][3.6][4.7] [4.10][6M] [7.1][7.3][7.4] [7.5][7.6][7.7]	[5.3] [5.5] [5.7]	[5.2] [5.3]	[5.2] [5.3] [5.8] [7.7]	[7.1] [7.3] [7.5] [7.6] [7.7]	[7.7]	[7.5]
LOW TO MEDIUM 1,000 TO 10,000	[3.2][3.3] [3.5][3.6] [3.7][4.11] [6A][7.2]	[3.2][3.3][3.5] [3.7][4.1][4.3] [4.10][4.11] [6A][7.2][7.3] [7.4][7.5]	[3.2][3.5] [3.7][4.1][4.4] [4.11][6A][7.2] [7.3][7.4][7.5]	[3.2][3.5][3.7] [4.1][4.3][4.7] [4.10][4.11][6A] [7.2][7.3] [7.4][7.5]	[3.2][3.3][3.5] [3.8][4.1][4.3] [4.10][4.11][6A] [7.2][7.3][7.4]	[3.2][3.3][3.5] [3.8][4.1][4.3] [4.7][4.10][4.11] [6A][7.3][7.4] [7.5]	[3.3][3.6][3.8] [4.1][4.3][4.4] [4.10][6A][7.5]	[3.3][3.8] [4.3][4.10] [6A][7.5]	[3.3][3.8] [4.3][4.10]	[3.3][3.8] [4.3][4.10]	[3.2][3.3][3.5] [3.7][4.1][4.3] [4.10][4.11][6A] [7.2][7.3][7.4] [7.5]	[4.1][4.7] [4.10][4.11] [6A][7.2][7.3] [7.4][7.5]	[5.3] [5.5] [5.6] [5.7]	[5.2] [5.3] [5.4]	[5.1] [5.2] [5.3]	[7.2] [7.3] [7.4] [7.5]		[7.5]
MEDIUM TO HIGH 10,000 TO 100,000	[3.2] [3.3] [4.11] [6A]	[3.9][4.1] [4.3][4.4] [4.5][4.11] [4.12][4.13] [6A][7.2][7.5]	[4.1][4.4] [4.5][4.11] [4.12][4.13] [6A][7.2]	[3.9][4.1][4.3] [4.4][4.5][4.11] [4.12][4.13] [6A]	[3.2][3.4][3.9] [4.1][4.3][4.4] [4.5][4.11] [4.12][4.13][6A]	[3.2][3.3][3.4] [3.9][4.1][4.3] [4.4][4.5][4.11] [4.12][4.13] [6A][7.5]	[3.3][3.4] [4.1][4.3] [4.4][4.5] [4.13][6A]	[3.3][3.4] [4.3][4.4] [4.5][4.13] [6A]	[3.3][3.4] [4.3][4.4] [4.13]	[3.3][3.4] [4.3][4.4] [4.5][4.13] [6A]	[3.1][3.3][3.4] [3.5][4.11] [4.12][4.13] [6A][7.2][7.5]	[4.1][4.4] [4.11][4.12] [4.13][6A] [7.2][7.5]	[5.1] [5.3] [5.5] [5.6] [5.10]	[5.1] [5.3] [5.10]	[5.1] [5.3] [5.9]	[4.11] [4.12]	[4.12] [4.13]	[4.5] [4.12]
HIGH 100,000+	[3.2] [3.3] [4.11] [6A]	[3.9][4.1] [4.2][4.3] [4.4][4.5] [4.12] [4.13][6A]	[4.12] [6A]	[3.9][4.2] [4.3][4.12] [6A]	[3.2][3.9][4.1] [4.2][4.3][4.4] [4.5][4.7][4.8] [4.11][4.12] [4.13][6A]	[3.2][3.3][3.4] [3.9][4.1][4.2] [4.3][4.4][4.5] [4.8][4.12][4.13] [6A]	[3.3][3.4] [4.1][4.3] [4.4][4.8] [4.13][6A]	[3.4][4.2] [4.3][4.4] [4.5][6A]	[3.4][4.3] [4.4][6A]	[3.4][4.2] [4.3][4.4] [6A]	[4.2][4.3] [4.12][6A]	[4.12] [6A]	[5.1] [5.6] [5.10]	[5.1] [5.3] [5.4] [5.10]	[5.9]	[4.7] [4.11] [4.12]	[4.12]	[4.5] [4.12]
ALL QUANTITIES	[3.1]	[3.1][3.6] [4.6][4.8] [4.9]	[3.6][4.6]	[3.1][3.6] [4.6][4.8] [4.9]	[3.1][3.6] [4.6][4.8] [4.9][7.5]	[3.1][3.6] [4.6][4.8] [4.9]	[3.1][4.6] [4.8][4.9]	[4.6][4.8] [4.9]		[4.6]	[3.1][3.6] [4.6][4.8] [4.9]	[4.8][4.9]				[7.5]	[3.6]	[3.6]

3.X Casting Processes / 4.X Forming Processes / 5.X Plastic & Composite Processing / 6.X Machining Processes / 7.X Non-traditional Machining Processes